

Mollification by a normalized smooth kernel

MathlibAnnex.Mollification.mollify

def

Defines convolution with a nonnegative smooth kernel of integral one.

Statement

Let $X = \mathbb{R}^n$ and $Y = \mathbb{R}^N$, with their coordinate sup norms and Lebesgue measure on X . For a real parameter ε and a map $f : X \rightarrow Y$, define $f_\varepsilon = \rho_\varepsilon * f$, where ρ_ε is the normalized bump described below.

Definition

For $\varepsilon > 0$, take the specified smooth bump φ_ε centered at 0, with inner radius $\varepsilon/2$ and outer radius ε . For $\varepsilon \leq 0$, the definition instead uses radii $1/2$ and 1 . Set

$$\rho_\varepsilon(y) = \frac{\varphi_\varepsilon(y)}{\int_{\mathbb{R}^n} \varphi_\varepsilon(z) dz},$$

$$f_\varepsilon(x) = \int_{\mathbb{R}^n} \rho_\varepsilon(y) f(x - y) dy.$$

The denominator is positive. This normalization followed by convolution is the complete definition. The fallback defines the operation at every real parameter; approximation uses only $\varepsilon \downarrow 0$.

Assumptions

The definition accepts every $n, N \geq 0$, every real ε , and every function f . It does not assume continuity, measurability, or integrability. Analytic properties of the convolution require their own hypotheses.

Conclusion

The output is the function $f_\varepsilon : X \rightarrow Y$. For locally integrable f this is the ordinary convolution with the compactly supported continuous kernel. For arbitrary f the displayed integral is understood with the source library's total Bochner-integral convention (zero when Bochner integrability fails).

For $\varepsilon > 0$, the same normalized bump satisfies

$$\rho_\varepsilon \geq 0, \quad \int_{\mathbb{R}^n} \rho_\varepsilon(y) dy = 1, \quad \rho_\varepsilon(y) = 0 \quad \text{if } \|y\|_\infty \geq \varepsilon.$$

In particular its topological support, $\overline{\{y : \rho_\varepsilon(y) \neq 0\}}$, is contained in $\overline{B}_\infty(0, \varepsilon)$. The definition fixes these bumps; no formula identifying them as dilates of a single kernel is asserted. Smoothness of f_ε and convergence are separate results.

Main citations

- Exact declaration and its source — `MathlibAnnex.Mollification.mollify`
- Normalized bump with positive-radius and fallback branches — `MathlibAnnex.Mollification.standardMollifier`
- Compact support of the normalized kernel — `MathlibAnnex.Mollification.hasCompactSupport_standardMollifier`
- Continuity of the normalized kernel — `MathlibAnnex.Mollification.continuous_standardMollifier`

Lean source signature (exact)

```
noncomputable def mollify {n N : ℕ} (ε : ℝ)
  (f : (Fin n → ℝ) → (Fin N → ℝ)) : (Fin n → ℝ) → (Fin N → ℝ) :=
  standardMollifier ε ★[ContinuousLinearMap.lsmul ℝ ℝ, volume] f
```

In the source	Mathematical meaning
<code>{n N : ℕ}</code>	The domain and output dimensions of $X = \mathbb{R}^n$ and $Y = \mathbb{R}^N$, both with coordinate sup norms; either may be zero.
<code>(ε : ℝ) (f : (Fin n → ℝ) → (Fin N → ℝ))</code>	The real parameter ε and arbitrary map $f : X \rightarrow Y$. This definition imposes neither $\varepsilon > 0$ nor regularity or integrability of f .
<code>standardMollifier ε</code>	The normalized bump ρ_ε . The cited definition uses radii $\varepsilon/2, \varepsilon$ for $\varepsilon > 0$, and $1/2, 1$ otherwise.
<code>ContinuousLinearMap.lsmul ℝ ℝ, volume</code>	Scalar multiplication of a vector in Y by a real number, and Lebesgue measure on X .

In the source	Mathematical meaning
<code>standardMollifier ε ★[ContinuousLinearMap.lsmul ℝ ℝ, volume] f</code>	The entire RHS: convolution $f_\varepsilon(x) = \int_X \rho_\varepsilon(y) f(x - y) dy$. It uses the total Bochner integral, which is zero when Bochner integrability fails.
<code>: (Fin n → ℝ) → (Fin N → ℝ)</code>	The output is the function $f_\varepsilon : X \rightarrow Y$. Approximation results require their own hypotheses.

▼ Exact surrounding binder context (separate excerpt)

```
noncomputable section
open Set MeasureTheory Filter
open scoped Convolution Topology ENNReal NNReal Pointwise
namespace MathlibAnnex
namespace Mollification
```

▼ **Exact content identity**

Declaration: MathlibAnnex.Mollification.mollify

Accepted content SHA-256: b4e9e1dfacea191863548272df300ed393323e9048f55d7f6a64aa6e32cb4b9

Accepted source guide SHA-256: c17d8730519abe8c478ed14b412685fe5ef985644d950fc028c307151ec31063

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73